

Evaluating Real-Time Object Detection Models for Autonomous Vehicle Vision

Surbhi Bahri

*School of Electrical Engineering and
Computer Science
University of Ottawa
Ottawa, Canada
sbahr036@uottawa.ca*

Hussein T. Mouftah

*School of Electrical Engineering and
Computer Science
University of Ottawa
Ottawa, Canada
mouftah@uottawa.ca*

Abstract—Object detection has made remarkable advancement in recent years particularly through CNN-based algorithms, essential for autonomous vehicles. These vehicles need efficient detection systems that balance speed and accuracy in complex environments. The study evaluates several real-time object detection models, such as YOLOv10, YOLOv11, EfficientDet, Faster R-CNN, and SSD focusing on both single-stage and two-stage detectors. Using the Berkeley DeepDrive 100K image dataset and iRain dataset, the study assesses the performance of these models in identifying key road-related objects. Additionally, the paper introduces an enhanced object detection system designed to improve robustness and accuracy in challenging conditions. The results emphasize the crucial role these advanced models have in improving the environmental awareness of autonomous vehicles, leading to better safety and operational performance on the road. Experimental data demonstrate that the proposed system surpasses current models, providing a scalable and dependable solution to boost the safety and efficiency of autonomous systems in real-world settings.

Keywords— Autonomous Vehicles, EfficientDet, Single Shot Multibox Detector, Object Detection, Convolutional Neural Network.

I. INTRODUCTION

Computer Vision (CV) is a specialized subfield within artificial intelligence (AI) that equips machines with the capacity to interpret and understand visual data, closely copying human visual perception. The ambition of computer vision is to develop algorithms and systems competent of examining and extracting significant information from images or video. Tasks such as object recognition, face detection, motion tracking, image segmentation, and comprehensive scene analysis are all part of this field. Ultimately, computer vision intends to qualify machines to make educated conclusions or take actions based on the visual input they gather [1, 21].

One of the fundamental challenges in computer vision is object detection, which involves not only classifying objects within an image but also locating them with exactness. This task demands algorithms that can both classify objects and uncover their positions through bounding boxes or segmentation, which is vital for numerous applications, specifically in autonomous systems. In recent years, deep learning has played a transformative part in enhancing computer vision abilities [8]. Convolutional Neural Networks (CNNs) have led to extensive advancements in the accuracy and efficiency of computer vision tasks. Former practices struggled to handle multifaceted and dynamic

environments, but CNNs and other deep learning models outshine at learning hierarchical features from large datasets, which enhances performance in tasks like object detection, image classification, and facial recognition. The expanding demand for real-time object detection in disciplines such as autonomous driving, robotics, and surveillance has directed the expansion of countless advanced models [18]. Distinguished architectures like YOLO (You Only Look Once), Faster R-CNN, SSD (Single Shot Multibox Detector), and EfficientDet have each presented unique approaches to confront the challenges of detecting objects speedily and precisely in dynamic environments.

YOLO has emerged as a key player in real-time object detection due to its novel approach of handling object detection as a single regression challenge. Instead of processing the image through distinct steps for classification and localization, YOLO foretells both the class and the bounding box for multiple objects in one unified pass. This outcomes in speedier processing times, making YOLO exceptionally appropriate for real-time applications. The newer versions, YOLOv10 and YOLOv11, assemble on prior repetitions by improving the network architecture to handle more intricate scenes with better precision. These versions emphasis on enhancing detection across a variety of object sizes and upgrading overall sturdiness, warranting high performance even in demanding conditions [1, 14].

Faster R-CNN is another prominent model, which extends the original R-CNN framework by encompassing a Region Proposal Network (RPN). The RPN permits the network to produce region proposals directly from the input image, bypassing the need for external region proposal algorithms. This speeds up the overall detection process while upholding accuracy, making Faster R-CNN well-suited for duties commanding precise object localization. While generally slower than models like YOLO, Faster R-CNN do extremely well in situations where the highest level of accuracy is essential [19]. On the other hand, SSD (Single Shot Multibox Detector) extends a explanation that balances both speed and accuracy. Like YOLO, SSD processes the image in a single pass, but it integrates multi-scale feature maps to enrich detection performance for objects of varying sizes. This makes SSD an attractive preference for applications where a combination of speed and precision is

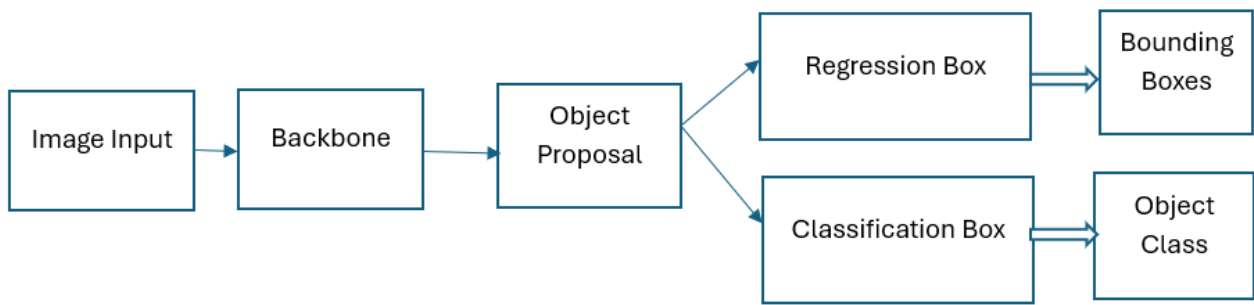


Fig. 1. An elementary two-stage Object Detector

obliged, and where objects in the scene may appear at distinct scales [5].

EfficientDet brings an innovative methodology to object detection by concentrating on optimizing model productivity. It utilizes a compound scaling scheme to scale the network's depth, width, and resolution concurrently, aiding high accuracy while managing fewer computational resources compared to other models. This makes EfficientDet particularly appreciated for applications in hampered environments, such as on embedded systems or devices with curbed computational power, where the balance between performance and resource utilization is critical [4].

Collectively, these models—YOLOv10, YOLOv11, Faster R-CNN, SSD, and EfficientDet—each contribute exclusive strengths to the object detection domain. While YOLO is benefited for its speed, Faster R-CNN shines in accuracy, and SSD strikes a middle ground. EfficientDet, with its prominence on computational efficiency, is devised for situations where resources are regulated. As the requirement for real-time, high-performance object detection continues to increase, these models are at the forefront of evolving technologies in autonomous systems and other AI-driven applications [9, 13].

In autonomous systems like self-driving cars, drones, and robots, precise and real-time object detection is fundamental for secure and efficient operation. These systems must detect and classify diverse objects—such as pedestrians, vehicles, traffic signals, and road obstacles—precisely and promptly. The capability to do so confirms that autonomous vehicles can navigate and make conclusions with a high degree of safety and dependability [8]. Furthermore, autonomous vehicles must function within dense and unpredictable environments, where accurate object detection is crucial for preserving situational cognizance. As developments in deep learning continue to augment these systems, the precision and speed of object detection will enhance, leading to enormous safety and reliability in self-driving vehicles. Consequently, object detection is a crucial technology driving advancement in autonomous vehicles, robotics, and

other intelligent systems [13]. This paper investigates how computer vision, strengthened by deep learning techniques, is enriching object detection for autonomous applications. By evaluating several state-of-the-art object detection models, we aim to assess their performance in real-world scenarios and grant discernments into how these technologies can be optimized for imminent autonomous systems. In conclusion, this paper extends an exhaustive assessment of the abovementioned architectures, evaluates their performance, and investigates potential future advancements.

II. ANALYSIS OF EXISTING LITERATURE

Object detection has become a foundation stone in many computer vision applications, notably in autonomous vehicles, robotics, and surveillance, where real-time detection of objects is essential. Plentiful models have emerged to address the contrasting challenges correlated with this assignment, each suggesting a unique set of strong suit and limitations [8]. This literature survey scrutinizes some of the most extensively adopted object detection architectures, comprising YOLOv10, YOLOv11, SSD (Single Shot Multibox Detector), EfficientDet, and Faster R-CNN, and discovers how these models contribute to progressing real-time detection proficiencies in dynamic environments.

The YOLO (You Only Look Once) series has played a fundamental responsibility in restructuring the panorama of object detection by establishing a one-stage methodology that is both rapid and efficient. Unlike customary two-stage techniques as shown in Fig. 1, where object proposals are first generated and then classified, YOLO streamlines the detection pipeline by envisaging both bounding boxes and class labels simultaneously in a single pass. This architecture facilitates speedier processing, making YOLO extremely appropriate for time-sensitive applications, such as autonomous vehicles [3, 10]. YOLOv10 and YOLOv11 further enhance upon previous editions with essential updates. YOLOv10 incorporates a transformer-based module for excellent universal feature extraction and employs a one-to-one matching system that eliminates the demand for

conventional post-processing techniques like Non-Maximum Suppression (NMS). Moreover, YOLOv10 pioneers a rank-guided proposal that augments convolution functions, improving its talent to detect objects precisely and promptly. YOLOv11 builds upon these improvements, fine-tuning the model for superior performance, mainly in intricate detection situations. Consequently, YOLO remains to dominate real-time object detection, specifically in applications commanding swift and trustworthy decision-making [21, 25].

The SSD (Single Shot Multibox Detector), introduced by Liu et al., also follows a one-stage approach to object detection, extending a balance between speed and accuracy. By producing object predictions at several feature scales, SSD is capable of detecting objects of different sizes successfully. This makes SSD a adaptable model, specifically in applications where varying object sizes ought to be detected in real-time. Although SSD is speedier than two-stage models like Faster R-CNN, it can struggle with detecting small or occluded objects in complex environments [5]. Despite this, SSD is an appealing selection for real-time applications where speed is prioritized over absolute accuracy, such as in video surveillance or autonomous driving. The adeptness of SSD makes it an alluring preference when rapid processing is necessitated for speedy decision-making, even if it loses some accuracy compared to more computationally demanding approaches [18].

EfficientDet, a model introduced by Tan et al. in 2020, focuses on providing high computational efficiency while sustaining scalability. Part of the EfficientNet family, EfficientDet utilizes a compound scaling method that simultaneously regulates the depth, width, and resolution of the network. This methodology permits the model to achieve a balance between accuracy and resource usage, making it exceptionally appropriate for devices with controlled computational power. EfficientDet also employs Bidirectional Feature Pyramid Networks (BiFPN) to enrich the blend of multi-scale characteristics, enriching its capability to detect objects across different sizes. EfficientDet has gathered substantial consideration in real-time applications that require both high performance and low resource utilization, such as in embedded systems for autonomous vehicles. Its capability to deliver state-of-the-art accuracy with limited parameters has made it a widely held selection for deployment in resource-constrained environments [4, 18].

Faster R-CNN, one of the earliest and most accurate two-stage models, employs a Region Proposal Network (RPN) to produce region proposals, chased by a second stage to

classify and polish these proposals. While Faster R-CNN is highly accurate, principally in detecting small or occluded objects, its two-stage process makes it slower than one-stage models like YOLO and SSD. This high computational cost

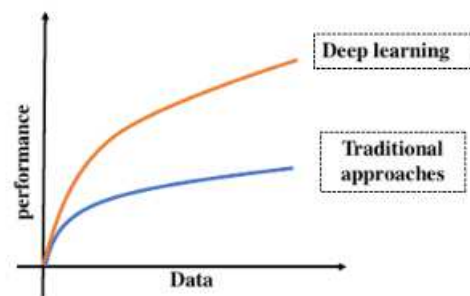


Fig. 2. Performance graph of traditional and DL approaches

restricts its application in real-time systems, such as autonomous driving, where rapidity is crucial. Nevertheless, Faster R-CNN endures an exceptional selection for applications where detection precision is more significant than processing speed, such as in high-resolution image assessment or medical imaging, where precision is paramount [6, 19].

In the context of autonomous vehicles, models that can handle multiple detection tasks simultaneously have become progressively worthy. YOLOP (You Only Look Once for Perception) is one such model devised explicitly for autonomous driving. YOLOP unites a mutual encoder with multiple decoders, each of which is accountable for detecting different road-related elements, such as lane markings, road signs, and pedestrians. This multi-task architecture permits YOLOP to perform comprehensive perception tasks, making it a precious model for autonomous vehicle applications. However, the added difficulty of conducting multiple tasks in parallel can lead to slower processing speeds compared to simpler single-task models like YOLO. Despite this, YOLOP's capability to simultaneously address various detection challenges in one framework makes it an operational tool for autonomous driving systems [7, 8, 9].

Additionally, the demand for lightweight models that can operate on mobile devices and edge systems has steered the growth of resourceful models like MobileNet and EfficientNet. EfficientNet, established in 2019, utilizes a novel compound scaling technique to advance model efficiency by regulating depth, width, and resolution simultaneously, ensuing in a more computationally efficient model. The architecture is aimed to uphold high accuracy while using limited resources, making it well-suited for environments with restricted computational competence. EfficientDet, an expansion of EfficientNet, further enhances the architecture for object detection chores, achieving high performance with insufficient parameters. This model has reaped considerable awareness for its capability to provide state-of-the-art accurateness while being lightweight enough for deployment in embedded systems, such as those observed in autonomous vehicles or portable devices [4, 16, 18].

The panorama of object detection has progressed considerably with the development of models like YOLOv10, YOLOv11, SSD, EfficientDet, and Faster R-

CNN. Each of these models carries exceptional strengths and trade-offs to the table, making them appropriate for different classes of applications. YOLO continues the go-to solution for real-time detection due to its speed and proficiency, with YOLOv10 and YOLOv11 extending additional advancements in performance [2, 25]. SSD is a fast and efficient model that is ideal for real-time detection, whereas EfficientDet do extremely well in resource-constrained environments by supplying high performance with limited parameters. Faster R-CNN in [19, 22] presents the greatest accuracy but at the cost of speed, making it better suited for applications where precision is more eminent than real-time processing. As these models persist to progress, forthcoming developments will possibly promote the frontiers of what is doable in real-time object detection, specifically for autonomous systems where both speed and accuracy are dire.

III. EXAMINED ALGORITHMS

Faster R-CNN is a extensively utilized two-stage object detection architecture acknowledged for its extraordinary accuracy. The model works in two stages: first, the Region Proposal Network (RPN) produces potential object regions, and second, the RoI pooling step enhances these regions to produce more accurate bounding boxes and classifications [6]. While the two-stage methodology grants highly precise object localization and classification, it can be slower compared to one-stage detectors due to the supplementary region proposal stage. This makes Faster R-CNN more appropriate for tasks where accuracy is ranked over speed, such as in applications necessitating exact detection of multifaceted objects, though it may not be superlative for real-time systems like autonomous vehicles that involve quick decision-making[19].

The SSD model takes a distinctive approach, choosing for a one-stage design that directly envisages both bounding boxes and object classes, eliminating the necessity for region proposals. Using a base network such as VGG16, MobileNet, or ResNet, SSD extracts feature and employs default bounding boxes of different aspect ratios and sizes to detect objects in an image. One of SSD's foremost strengths is its capacity to handle multi-scale object detection by employing the model at different feature map resolutions [9]. This makes it skilled at detecting both large and small objects, and the one-stage design guarantees speedier inference times compared to two-stage models like Faster R-CNN. SSD is well-suited for real-time applications, such as autonomous driving, where both speed and accuracy are essential [5].

EfficientDet is a lightweight, still prevailing object detection model constructed with scalability in concentration, intensely appropriate for environments where computational resources are regulated. It leverages EfficientNet, a favorably efficient image classification model, and incorporates the BiFPN (Bidirectional Feature Pyramid Network) for efficient multi-scale feature fusion [13]. This combination certifies that the model can

effectively acquire notable features from different scales, while still preserving minimal computational cost, which is ideal for resource-constrained applications. The BiFPN module is vital in EfficientDet, granting bidirectional feature flow to enrich the fusion of features across several levels of the network. The model's compound scaling method strikes an admirable balance between accuracy and computational petition, making it principally well-suited for applications in autonomous vehicles, robotics, and other real-time systems where low latency is essential but resource usage is limited [4].

YOLOv10 develops upon its predecessors by proposing numerous innovative design amendments, such as a transformer-based module that augments the extraction of global image features. YOLOv10 eliminates the requirement for post-processing like NMS by employing a one-to-one matching mechanism, which directly outputs object detections [2]. Major enhancements in YOLOv10 involve the introduction of CSPNet and Partial Self-Attention (PSA) modules, which facilitate boost the feature extraction process while sustaining efficiency. These advancements acknowledge YOLOv10 to accomplish high performance with lower computational costs, making it appropriate for real-time applications in systems like autonomous driving [21].

YOLOv11 encourages on the enhancements instituted in former YOLO versions, concentrating on boosting both detection accuracy and processing speed. It incorporates a number of valuable architectural novelties, such as the C3K2 block, SPFF (Spatial Pyramid Pooling Fast) module, and C2PSA block (Cross Stage Partial with Spatial Attention). These enrichments facilitate YOLOv11 to better capture spatial information and enhance feature representation, remarkably for small or occluded objects. YOLOv11's backbone starts with a Convolutional Block, supported by Bottle Neck layers. It also pioneers the C2F (Cross Stage Partial Focus) block, which divides the feature map into two parts and processes them independently to boost feature extraction. The C3K2 block is fundamental to YOLOv11, employing 3x3 convolutions to process the feature map competently. This design permits the model to capture necessary features at assorted stages of the backbone while guarding computational overhead low. The SPFF module pools features across various scales, making it simpler to detect objects of different sizes. This feature is specifically cooperative for distinguishing small objects, which have conventionally been a challenge for earlier YOLO versions. The C2PSA block proposes attention mechanisms to assist the model emphasize on applicable regions within the image, specifically for smaller or partially occluded objects. By highlighting spatial information, the model can detect fine-grained aspects with higher accuracy. In the final stage, YOLOv11 employs a multi-scale projection head that outputs detections at distinctive levels of granularity,

permitting the model to detect both small and large objects across numerous feature maps [25].

IV. ENVIRONMENTAL TESTING

The enhanced object detection model, after undergoing validation, is deployed and tested in practical settings such as autonomous driving, robotics, and surveillance. During deployment, the model is integrated with hardware systems that capture real-time images and video feeds. This phase focuses on ensuring the model operates efficiently within the constraints of available computational resources like GPUs or edge devices. Key considerations include optimizing the model for hardware limitations, evaluating real-time performance with minimal delay, and testing its effectiveness in varying environmental conditions, such as different weather patterns, lighting, and traffic scenarios. Field testing is conducted to verify that the model maintains accuracy and speed, which is crucial for applications requiring rapid decision-making and operational reliability [27].

To better understand the model's response to different weather conditions, synthetic rain is introduced into clear-weather images for comparison. This method allows testing the model under controlled conditions to see how detection accuracy changes in different weather scenarios. The model is also tested with a "test rainy set" of images taken during actual rainy conditions to further evaluate its performance. The system's potential is significant across various fields: it enhances autonomous driving by improving detection accuracy in challenging weather, supports better robotic navigation in industrial and logistics tasks, and strengthens surveillance operations in adverse weather. The system's ability to adapt to real-world conditions makes it highly suitable for applications requiring reliability and performance under varying environmental factors [28].

V. EXPERIMENTAL SETUP

In this experimentation, each model was trained independently using an appropriate GPU environment on the Berkeley DeepDrive 100K and iRain dataset, which is broadly used for autonomous vehicle vision tasks. Once training was finalized, the models were transformed to TensorFlow Lite (TFLite) format. TensorFlow grants a vigorous collection of tools, models, and functions intended to streamline the deep learning pipeline. Explicitly, we employed the TensorFlow Lite Model Maker library, which proposes a suitable way to convert pretrained models like EfficientDet, SSD, Faster R-CNN, and YOLO into .tflite format. These TFLite models can then be saved, stored and employed for real-time applications in resource-limited environments. The GPU provision was vital in fast-tracking the training process, permitting for faster model development, evaluation, and conversion into the efficient TFLite format, eventually augmenting the swiftness and effectiveness of arraying these models for autonomous driving systems [26].

VI. RESULTS

This study evaluated the performance of five popular object detection models as shown in Table I. — YOLOv10, YOLOv11, EfficientDet, Faster R-CNN, and SSD — based

TABLE I. MODEL COMPARISON RESULTS

Model	Berkeley DeepDrive test set (mAP)	iRain test set (mAP)	Object Detection Approach
Faster R-CNN	82.34	78.22	Two-Stage
SSD	73.82	67.94	Single-stage
EfficientDet	77.12	73.85	Single-stage
YOLOv10	75.23	73.26	Single-stage
YOLOv11	80.21	78.87	Single-stage

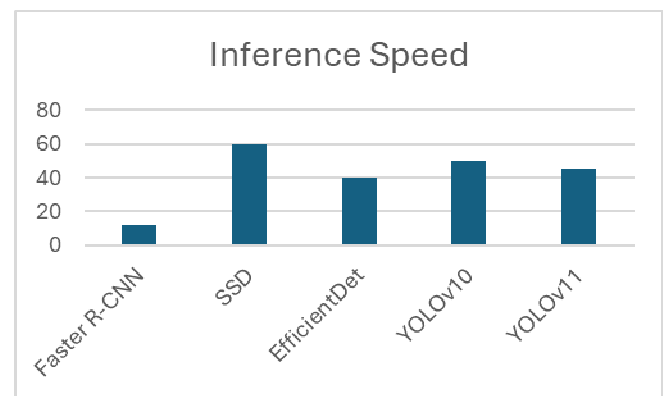


Fig. 3. Inference Speed evaluation results of different Object Detection models

on two essential metrics: Mean Average Precision (mAP) and inference speed using the Berkeley DeepDrive test set and the iRain test set. Faster R-CNN, a two-stage detection model, achieves the highest mAP score of 82.34 on the Berkeley test set but experiences a slight decrease to 78.22 on the iRain test set. Despite its strong accuracy, Faster R-CNN's slower inference speed of 12 FPS makes it less suitable for real-time applications that require fast processing. In contrast, SSD, a single-stage model, delivers faster performance with 60 FPS but has lower accuracy, with mAP scores of 73.82 on the Berkeley set and 67.94 on the iRain set. EfficientDet shows a balance between speed and accuracy, achieving a mAP of 77.12 on the Berkeley set and 73.85 on the iRain set, while operating at 40 FPS, making it an efficient choice without compromising much on precision. YOLOv10 and YOLOv11, both single-stage models, also demonstrate competitive performance, with YOLOv11 leading the group with an mAP of 80.21 on the Berkeley test set and 78.87 on the iRain set. While YOLOv10 operates at a speed of 50 FPS, YOLOv11 provides a slightly lower frame rate of 45 FPS, but with improved detection capabilities. These results indicate that although two-stage models like Faster R-CNN typically offer better accuracy, especially under ideal conditions, single-stage models like YOLOv11 strike a better balance between performance and speed. The

drop in mAP scores on the iRain test set for all models illustrates how weather conditions can significantly affect detection accuracy. This is an important consideration for deploying these models in real-world environments where both speed and accuracy are essential for reliable performance.

VII. CONCLUSIONS

This study conducted an in-depth assessment of five broadly recognized object detection models—YOLOv10, YOLOv11, EfficientDet, Faster R-CNN, and SSD—concentrating on their Mean Average Precision (mAP) and inference speed. The discoveries indicated that Faster R-CNN accomplished the highest accuracy but was slower in processing, which can curb its use in real-time applications. In contrast, YOLOv10 and YOLOv11 bid a reasonable trade-off between speed and accuracy, with YOLOv11 outperforming YOLOv10 in terms of both metrics. EfficientDet was found to be an outstanding preference for environments with limited resources, striking a balance between performance and computational efficiency. SSD, although the rapidest, flashed marginally reduced accuracy, making it more appropriate for applications where speed is ranked over exactness. This study highlights the significance of choosing a model designed to the precise needs of autonomous systems and real-time application.

VIII. FUTURE DIRECTIONS

Future improvements in object detection systems for autonomous vehicles and robotics present significant opportunities. Enhancing domain adaptation techniques to handle challenging weather conditions, such as heavy rain, snow, or fog, will be crucial. Advanced models like transformers or graph neural networks could further elevate detection and tracking performance. Additionally, increasing scalability and flexibility to function across diverse environments, including rural areas, is essential for broader system deployment. Integrating IoT devices could provide additional data sources, improving overall accuracy and decision-making. Expanding into advanced applications, such as autonomous drones and reinforcement learning, would foster smarter, more autonomous decision-making. Further optimization of models through pruning, quantization, and distillation techniques could reduce computational load while maintaining or enhancing performance. Research into hybrid models that combine the advantages of both one-stage and two-stage detectors could offer an optimal balance between speed and accuracy. Evaluating these models in real-world environments, including varied weather and low-light conditions, will be essential for assessing their robustness. Additionally, focusing on edge-device deployment and minimizing latency will ensure real-time efficiency, while transfer learning and domain adaptation can further enhance model specialization for specific tasks, boosting their effectiveness in critical applications like autonomous driving [2, 7, 8].

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